In this paper, all excercises in Hoffman / Kunze will be solved in Complete form.

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ A_{n1} & \cdots & \cdots & A_{nn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ B_{n1} & \cdots & \cdots & B_{nn} \end{pmatrix}$$

Denote $e_j \in F^{n\times 1}$, then

$$Ae_j = \begin{pmatrix} A_{1j} \\ A_{2j} \\ \vdots \\ A_{nj} \end{pmatrix} \quad \text{and} \quad e_i^T A = \begin{pmatrix} A_{i1} & A_{i2} & \cdots & A_{in} \end{pmatrix}$$

$$AB = \left(\sum_{k=1}^{n} A_{ik} \cdot B_{kj} \right)_{i,j} = \left(f(A^T e_i, Ae_j) \right)_{i,j},$$

where $f : F^n \times F^n \to F : \left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto \sum_{k=1}^{n} x_k y_k$, which is the bilinear form on F^n.

[Sec 3.1. Linear Transformation.

#3. Let $V = F[x]$, and $D : V \to V$ be a formal differentiation.
Find kernel and range of D.

Proof. If the characteristic of F is 0, then

$$Df = 0 \iff f \text{ is constant, thus } \ker D = F.$$

And, the basis for V is $\{x^k \mid k = 0, 1, 2, \cdots\}$, so $Dx^k = k \cdot x^{k-1}$ $(k \geq 1)$ implies

$$\operatorname{Im} D = V.$$

If the characterstic of F is not zero, a prime p, then

$$Dx^n = 0 \iff n \in (p)$$

$\ker D = \operatorname{Span}_F \{x^{pk} \mid k = 0, 1, 2, \cdots\}$

And

$\operatorname{Im} D = \operatorname{Span}_F \{x^{n-1} \mid n \in \mathbb{N} \text{ and } n \notin (p)\}$

#6. Describe explitly a linear operator $T : F^2 \to F^2$ s.t

$$T(e_1) = \begin{pmatrix} a \\ b \end{pmatrix} \text{ and } T(e_2) = \begin{pmatrix} c \\ d \end{pmatrix}$$

Proof. $[T]_{\{e_1, e_2\}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$.

#8. Find a linear $T : \mathbb{R}^3 \to \mathbb{R}^3$ s.t $\operatorname{Im} T = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$

Proof. Take $[T]_{\{e_1, e_2, e_3\}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ -1 & 2 & 2 \end{pmatrix}$.

#9. Verify for fixed $B \in M_{n\times n}(F)$,

$$T : M_{n\times n}(F) \to M_{n\times n}(F) ; A \mapsto AB - BA$$

is linear map.

proof. Given $A_1, A_2 \in M_{n\times n}(F)$,

$$T(A_1 + A_2) = (A_1 + A_2)B - B(A_1 + A_2)$$

$$= A_1 B - BA_1 + A_2 B - BA_2$$

$= T(A_1) + T(A_2)$. And clearly $T(cA) = c(AB - BA) = cT(A)$ for any $c \in F$.

#10. Find a map on $\mathbb{C}$ to $\mathbb{C}$ is $\mathbb{R}$-linear but not $\mathbb{C}$-linear.

proof. $T : \mathbb{C} \to \mathbb{C} ; a + bi \mapsto a$.

Given $a+bi, c+di \in \mathbb{C}$,

$$T(a+bi + c+di) = a+c = T(a+bi) + T(c+di),$$

So, it preserves the addition. And, for any real number $r \in \mathbb{R}$,

$$T(r \cdot a + r \cdot bi) = ra = r \cdot T(a+bi)$$

But for $i \in \mathbb{C}$,

$$T(i \cdot a + i \cdot bi) = T(-b + ai) = -b \neq i \cdot T(a+bi) = ia. \quad \blacksquare$$

#11. Let $A \in M_{m\times n}(F)$.
Prove that $T: F^n \to F^m ; X \mapsto AX$

is zero map if and only if A is zero matrix.

Proof. T is zero map $\Leftrightarrow$ for any $X \in F^n$, $T(X) = 0$

$\Leftrightarrow$ for each $1 \le i \le n$, $T(e_i) = 0$

$\Leftrightarrow$ 〃 $Ae_i = 0$

$\Leftrightarrow$ all rows of A are zero

$\Leftrightarrow$ $A \equiv 0$.

#13. Let $T: V \to V$ be a linear operator. Then,

$\operatorname{Im} T \cap \ker T = \{0\}$ if and only if $T(T(\alpha)) = 0 \Rightarrow T(\alpha) = 0$.

Proof. Since $T(\alpha) \in \operatorname{Im} T$ for all $\alpha \in V$, if $T(T(\alpha)) = 0 \Rightarrow T(\alpha) \in \ker T \Rightarrow T(\alpha) = 0$.
Conversely, for any $\beta \in \operatorname{Im} T \cap \ker T$, $\beta = T(\alpha)$ for some α, and, if $\beta = 0$ by the assumption.

「Sec 3.2.

#3. $T: \mathbb{R}^3 \to \mathbb{R}^3 ; (x_1, x_2, x_3) \mapsto (3x_1, x_1 - x_2, 2x_1 + x_2 + x_3)$

is invertible?

Prof $T(e_1) = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ $T(e_2) = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$ $T(e_3) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

So $[T]_\beta = \begin{pmatrix} 3 & 0 & 0 \\ 1 & -1 & 0 \\ 2 & 1 & 1 \end{pmatrix}$ is invertible, $\det T = -3$.

#6. Let $T: F^3 \to F^2$ and $U: F^2 \to F^3$ be linear maps

Prove that $U \circ T : F^3 \to F^3$ is not invertible

Proof. $U \circ T$ invertible $\Leftrightarrow$ $U \circ T$ is onto, by the dimension Theorem.

But, $U \circ T[F^3] = U[F^2] \neq F^3$, since $\dim F^2 = 2 < 3 = \dim F^3$.

#7. Find $T, U : \mathbb{R}^2 \to \mathbb{R}^2$ s.t. $TU = 0$ but $UT \neq 0$.

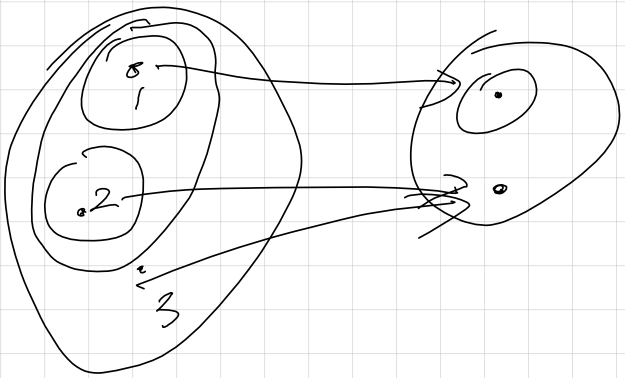
Prof. Let $T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

$TU = 0$, but $UT = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0$.

#8. Find $T^2 = 0$ but $T \neq 0$. $T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

#9. Let V be a fin. dim space, and $T, U : V \to V$ be linear operators s.t. $TU = I$.
Then, T is invertible with $T^{-1} = U$.
Find an Example it is false when V is not fin. dim.
Proof. Since $(T \circ U)[V] = V$, $U[V] = V$, so U is onto.
By the dimension Theorem, $\ker U = \{0\}$, so U is invertible.
Now, $T = U^{-1}$.

#11. Let V be a fin. dim vector space and $T: V \to V$ be a linear operator.
If $\operatorname{rank} T^2 = \operatorname{rank} T$, then $\ker T \cap \operatorname{Im} T = \{0\}$. (So $\ker T \oplus \operatorname{Im} T = V$).
Proof. Since for any $\alpha \in V$, $T(T(\alpha)) \in \operatorname{Im} T^2$, so $\operatorname{Im} T \subseteq \operatorname{Im} T^2$.
And, the dimension are same by assertion, $\operatorname{Im} T = \operatorname{Im} T^2$,
Moreover, if $T(\alpha) = 0$, then $T(T(\alpha)) = 0$, so $\ker T \subseteq \ker T^2$.
and, similarly, $\ker T = \ker T^2$.
Now, let $\beta \in \ker T \cap \operatorname{Im} T$. Then, $\beta = T(\alpha)$ for some $\alpha \in V$ and $T(\beta) = T^2(\alpha) = 0$.
$T^2(\alpha) = 0$ implies $\alpha \in \ker T^2$, so $\alpha \in \ker T$. That is, $\beta = T(\alpha) = 0$.
So proved.



$\{\{1\},\{2\},\{1,2\},X,\emptyset\}$ $\{\{1\},Y,\emptyset\}$

$[0,2\pi]\to\mathbb{C};\ t\mapsto e^{ti}$

$[0,\pi)\to X$

$\pi:\mathbb{R}^2\to\mathbb{R};\ (x,y)\mapsto x$

$\{\frac{1}{n},n\}$ $\{(\frac{1}{n},n)\mid n\in\mathbb{N}\}$

Sec 3.3. Let V and W be vector spaces over F, and $U: V \to W$ be an isomorphism.
Prove that

$$\mathcal{L}(V,V) \to \mathcal{L}(W,W) : T \mapsto UTU^{-1}$$

is an isomorphism.

「Sec. 3.5 linear functional.

#3.5.3. For any $A, B \in M_{n\times n}(F)$, $\operatorname{trace}(AB) = \operatorname{trace}(BA)$.
Moreover, if $A \sim B$, then $\operatorname{trace} A = \operatorname{trace} B$.

Proof. Since $\operatorname{trace} AB = \sum_{j=1}^{n}\sum_{k=1}^{n} A_{jk}B_{kj}$ and $\operatorname{trace} BA = \sum_{j=1}^{n}\sum_{k=1}^{n} B_{jk}A_{kj}$

so just rename the indices, we otain the first assertion.
Now, suppose that $A = U^{-1}BU$ for some $U \in GL(F)$. Now,

$$\operatorname{trace} A = \operatorname{trace} U^{-1}BU = \operatorname{trace} BUU^{-1} = \operatorname{trace} B.$$

Sec 6.2.

6.2.1. Denote f_i is the characteristic polynomial of the given Matrix A_i.

$f_1(t)=t(t-1)$, $f_2(t)=(2-t)(1-t)+3=t^2-3t+5$, $f_3(t)=(1-t)^2-1=t^2-2t$.

In $\mathbb{R}$, f_2 and $\cancel{f_3}$ do not split over the given field, actually there are No solution in $\mathbb{R}$ of f_2 ~~and f_3~~. So A_2 and $\cancel{A_3}$, has no eigenvector.

But, $\begin{bmatrix}1&0\\0&0\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix}=0$, so $E_0=\ker A=\operatorname{span}_{\mathbb{R}}\{\begin{pmatrix}0\\1\end{pmatrix}\}$

And $\begin{bmatrix}1&0\\0&0\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix}$, so $E_1=\operatorname{span}_{\mathbb{R}}\{\begin{pmatrix}1\\0\end{pmatrix}\}$, $\quad \frac{2\pm\sqrt{4-8}}{2}$

In $\mathbb{C}^2$, A_1 has same eigenvectors. But;

$$t=\frac{3\pm\sqrt{9-20}}{2}=\frac{3\pm i\cdot\sqrt{11}}{2}$$ are roots of f_2, and

$$t=\frac{2\pm\sqrt{4-8}}{2}=1\pm i$$ are roots of f_3. So,

$$A_2-\underbrace{\left(\frac{3\pm i\sqrt{11}}{2}\right)}_{\lambda_2}\cdot I=\begin{bmatrix}\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2} & 3\\ -1 & -\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2}\end{bmatrix}$$

Then, $$(A_2-\lambda_2)I\cdot\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\left(\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)x+3y\\ -x+\left(-\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$$

$$\iff \begin{cases}\left(\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)x+3y=0\\ \left(\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)x+3y=0.\end{cases}$$ (multiplying $\left(\frac{1}{2}-i\cdot\frac{\sqrt{11}}{2}\right)$ to the bottom)

So, the eigenvectors are:

$$\begin{pmatrix}-1\\ \frac{1}{3}\left(\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2}\right)\end{pmatrix},$$ corresponding to $\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2}$

#6.2.6. $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix}$ over $\mathbb{R}$.

$\det(A - tI) = t^4$, and A has the only eigenvalue as 0. So,

A is diagonalizable iff $\dim \operatorname{Ker} A = 4$.

But, the only possibility for $\dim \ker A = 4$ is $a = b = c = 0$,
because if at least one of a or b or c is not zero, then $\operatorname{rank} A \geq 1$.

#6.2.7. If $T: F^n \to F^n$ has n distinct eigenvalues, then T is diagonalizable.

proof Let $\alpha_1, \cdots, \alpha_n \in F^n$ be the eigenvectors corresponding to distinct eigenvalues.
Then, $\{\alpha_1, \cdots, \alpha_n\}$ are linearly independent, because: Suppose that

$$c_1\alpha_1 + c_2\alpha_2 + \cdots + c_n\alpha_n = 0 \quad \text{where } c_i \in F.$$

Then for any F-polynomial $f(t) \in F[t]$,

$$0 = f(T)(c_1\alpha_1 + \cdots + c_n\alpha_n) = f(c_1)\cdot\alpha_1 + \cdots + f(c_n)\alpha_n \cdots (*)$$

Take $f_i(t) = \prod_{j \neq i}(t - c_j) \Big/ \prod_{j \neq i}(c_i - c_j)$

Then, $f_i(c_j) = \delta_{ij}$, so, we can conclude $c_1 = c_2 = \cdots = c_n = 0$,

by substituting $f_1, \cdots, f_n$ to $(*)$.

#6.2.8. Let $A, B \in M_{n\times n}(F)$. If $I-AB$ is invertible, then $I-BA$ is invertible with

$$(I-BA)^{-1} = I + B\cdot(I-AB)^{-1}\cdot A$$

Proof. Let $C = (I-AB)^{-1}$. Then,

$$C - ABC = C - CAB = I, \text{ so } ABC = CAB \text{ (That is, } C \text{ and } AB \text{ commute.)}$$

Now $(I-BA)\cdot(I+BCA) = I + BCA - BA - BABCA$

$$= I + BCA - BA - B(C-I)A$$
$$= I + BCA - BA - BCA + BA$$
$$= I.$$

So, proved.

#6.2.9. Using the fact above, deduce that

AB and BA have precisely the same eigenvalue in F.

Proof. If $t=0$, then

t is an eigenvalue for $AB \iff \det AB = 0 \iff \det BA = 0 \iff t$ 〃 BA.

If $t \neq 0$, then $\det(AB - tI) = 0 \iff t^{-n}\cdot\det((AB)\cdot t^{-1} - I) = 0$

$$\iff \det((BA)\cdot t^{-1} - I) = 0$$
$$\iff \det(BA - tI) = 0.$$

So, proved.

#6.2.10. 2 by 2 Symmetric Matrix is diagonalizable.
Proof. Let $A=\begin{pmatrix} a & c \\ c & b \end{pmatrix}$ be given. Then,

$$\det(A-tI)=(t-a)(t-b)-c^2=t^2-(a+b)t+ab-c^2$$

If $c=0$, then it is clearly diagonal. And, if $c\neq 0$, then $c^2>0$, so the charateristic polynomial has two distinct roots, denote r_1 and r_2,

$$\dim \ker(A-r_1I)\geq 1 \quad \text{and} \quad \dim \ker(A-r_2I)\geq 1,$$

So $\dim\ker A-r_1I=\dim\ker A-r_2I=1$, so diagonalizable.

#6.2.11. If $N\in M_{2\times 2}(\mathbb{C})$ satisfies $N^2=0$, then either $N=0$ or $N\sim\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
Proof. If $N=0$, then we are done. Suppose that $N\neq 0$, but $N^2=0$.
(Directly, the minimal polynomial of N is t^2, so it can be triangular.
But in this, we prove more specifically.)
Since $N^2=0$ implies $N\cdot Nx=0$ for all $x\in\mathbb{C}^2$, so

$$\operatorname{Im} N\subseteq \ker N.$$

Thus $\ker N\neq\{0\}$, take $\alpha_1\in\ker N$ be non-zero.
And, since $N\neq 0$, $\ker N\neq\mathbb{C}^2$. So, take $\alpha_2\in\mathbb{C}\setminus\ker N$ s.t $N\alpha_2\in\ker N$.
Now, $N\alpha_1=0$ and $N\alpha_2=c\cdot\alpha_1$ for some non-zero scalar $c\in F$, so

$$[L_N]_{\{\alpha_1,\alpha_2\}}=\begin{bmatrix} 0 & c \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad [L_N]_{\{\alpha_2,\alpha_1\}}=\begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix},$$

(To conclude the conversation completely, take $\alpha_2'=c^{-1}\alpha_2$).

#6.2.12. Every complex 2 by 2 Matrix is similar to $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ or $\begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix}$.
Proof. The charateristic polynomial of the given matrix is must be degree 2 complex polynomial,
So, there are only two possibilites: the polynomial has only one root of multiplicity 2
or distinct two roots.
In the second case, it is diagonalizable. In the first case, let $c\in F$ be the root.
Then, $(N-cI)^2=0$, because $\det(N-tI)=(t-c)^2$, so $\ker N-cI\neq 0$.
So $N-cI\sim\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, or $N\sim\begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix}$.

#6.2.13.

Let $V = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ is continuous}\}$, and define

$$T : V \to V ; f \mapsto \int_0^x f(t)\, dt.$$

Then T has no eigenvalues.

Proof. Since the integral has linearity, T is a linear map.

Suppose that T has an eigenvalue, $c \in \mathbb{R}$. That is,

$$T(f) = c \cdot f(x) = \int_0^x f(t)\, dt \quad \text{for some non-zero } f \in V.$$

If $c = 0$, then $\int_0^x f(t)\, dt = 0$ for all x, so f is zero, contradiction.

If $c \neq 0$, then $c \cdot f(0) = \int_0^0 f = 0$. Moreover,

$$c \cdot F'(x) = F(x) \Rightarrow F(x) = c^{-1} e^{cx} \to f(x) = e^{cx}$$

But $e^{c0} = e^0 = 1 \neq 0$, so contradiction.

#6.2.14.

Let $A \in M_{n \times n}(F)$ be a matrix with ch. polynomial

$$(x - c_1)^{d_1} \cdots (x - c_k)^{d_k}$$

Let $V = \operatorname{span}_F \{B \in M_{n \times n} \mid AB = BA\}$, then $\dim V = d_1^2 + \cdots + d_k^2$.

Proof.

#6.3.2.

Let $a,b,c \in F$. Then,

$A = \begin{bmatrix} 0 & 0 & -c \\ 1 & 0 & -b \\ 0 & 1 & -a \end{bmatrix}$ has characteristic polynomial as x^3+ax^2+bx+c, and it is minimal polynomial for A.

Proof. $\det(xI-A) = \begin{vmatrix} x & 0 & c \\ -1 & x & b \\ 0 & -1 & x+a \end{vmatrix} = x\cdot(x(x+a)+b)+c = x^3+ax^2+bx+c.$

And,

$$A^2 = \begin{bmatrix} 0 & -c & ac \\ 0 & -b & -c+ab \\ 1 & -a & -b+a^2 \end{bmatrix} \text{ and } A^3 = \begin{bmatrix} -c & ac & bc-a^2c \\ -b & -c+ab & ac+b^2-a^2b \\ -a & -b+a^2 & -c+2ab-a^3 \end{bmatrix}$$

So, by the direct calculation, $A^3+aA^2+bA+cI = O$.

Now, $\{I, A, A^2\}$ is linearly independent, so there is no polynomial $p(x)$ of degree 2 s.t.

$$p(A)=0$$

So, the polynomial x^3+ax^2+bx+c is minimal polynomial for A.

#6.3.3. Put $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ -2 & -2 & 2 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix}$. Then, $\det(xI-A) =$

#6.3.5. Let V be an n-dimensional Vector Space and $T: V \to V$ be a linear operator. If $T^k = 0$ for some $k \in \mathbb{N}$, then $T^n = 0$.

Proof. If $k \leq n$, then clear, $T^n = T^{n-k} \circ T^k = 0$.
Suppose that $k > n$. By the Cayley-Hamilton Theorem, the minimal polynomial $p(x)$ of T satisfies $\deg p(x) \leq n$, since $p(x)$ divides the charictorstic polynomial of T.
So, the minimal polynomial of T also divides T^k, so $T^r = 0$ for some $r \leq n$.
Now, $T^n = T^{n-r} \circ T^r = 0$.

#6.3.6. Find a 3 by 3 Matrix for which the minimal polynomial is x^2.

Proof. A has minimal polynomial as x^2 if and only if:

$$A^2 = 0 \text{ and } \{A, I\} \text{ is linearly independent.}$$

So, $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

#6.3.7. Let $D: P_n(\mathbb{R}) \to P_n(\mathbb{R})$ be the differentiation operator. Evaluate the minimal polynomial for D.

Proof. For any Polynomial $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in P_n(\mathbb{R})$,

$$D^n\left(\sum a_i x^i\right) = n! \cdot a_n$$

And, so $D^{n+1} = 0$, the zero map. Now, suppose that

$$c_n D^n + c_{n-1} D^{n-1} + \cdots + c_1 D + c_0 I = 0, \quad \text{where } c_0, \cdots, c_n \in \mathbb{R}.$$

Substracting Polynomials;

$$f_0(x) = 1, \; f_1(x) = x, \; f_2(x) = \frac{1}{2!} \cdot x^2, \; f_3(x) = \frac{1}{3!} \cdot x^3, \; \cdots, \; f_n(x) = \frac{1}{n!} \cdot x^n$$

Then, the all coefficients are zero, so $\{I, D, \cdots, D^n\}$ is linearly independent.
In other say, the polynomial of degree less than $n+1$ cannot be the minimal polynomial.

+Comment for 6.3.7. $\mathbb{R}$이 아니라 일반적인 F 에서 가능, 하지만 $ch(F) = 0$ 이어야 함
더 약화해도 될것 같으나, 귀찮다.

#6.3.8.
Since $P^2 = P$, the minimal polynomial is $x^2 - x = x(x-1)$.

#6.3.9. If the characteristic polynomial of $A \in M_{n\times n}(F)$ is

$$f = (x-c_1)^{d_1} \cdots (x-c_k)^{d_k},$$

then trace $A = c_1 d_1 + \cdots + c_k d_k$.

Proof. Since $f = x^n - (c_1 d_1 + \cdots + c_k d_k) \cdot x^{n-1} + \cdots$,

therefore trace $A = c_1 d_1 + \cdots + c_k d_k$.

「Sec 6.4.

#6.4.1. Let $A=\begin{bmatrix}1 & -1\\ 2 & 2\end{bmatrix}\in M_{2\times 2}(\mathbb{R})$.

a). The L_A-invariant subspace are only $\mathbb{R}^2$ and $\{0\}$.

Proof. Let $\{0\}<W<\mathbb{R}^2$ be a L_A-invariant subspace.

Then, $W=\text{span}_{\mathbb{R}}(a,b)$ for some non-zero vector $(a,b)\in\mathbb{R}^2$. So, by the assumption,

$A\cdot\begin{pmatrix}a\\ b\end{pmatrix}\in W$, or $A\begin{pmatrix}a\\ b\end{pmatrix}=c\cdot\begin{pmatrix}a\\ b\end{pmatrix}$ for some $c\in\mathbb{R}$.

But, $\det(xI-A)=(x-1)(x-2)+2=x^2-3x+4=(x-\frac{3}{2})^2+\frac{7}{4}$ has no solution in $\mathbb{R}$, so contradiction.

b). If A is a matrix in $\mathbb{C}$, then it has an eigenvector α, so $\text{Span}_{\mathbb{C}}\{\alpha\}$ is we desired.

#6.4.2. Let W be an invariant space for T.

Then, the minimal polynomial of T_W divides the minimal polynomial of T.

Proof. Let $P(x)$ be the minimal polynomial of T. Then, for any $\alpha\in W$,

$$P(T)(\alpha)=P(T_W)(\alpha)=0.$$

So, $P(T_W)=0$, thus we obtain the result.

#6.4.3. Let $c\in F$ be an eigenvalue of T.

For all $\alpha\in\text{Ker}(T-cI)$, $T(\alpha)=c\cdot\alpha$, so

$$T|_{\text{Ker}(T-cI)}=c\cdot I:\text{Ker}(T-cI)\to\text{Ker}(T-cI):\alpha\mapsto c\alpha.$$

So, the restriction is c-multiple of Identity map if $c\neq 0$, or zero map if $c=0$.

#6.4.4.

Let $A=\begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$ over $\mathbb{R}$. Find triangularization for A.

$$\det(xI-A)=\begin{vmatrix} x & -1 & 0 \\ -2 & x+2 & -2 \\ -2 & 3 & x-2 \end{vmatrix} = x((x+2)(x-2)+6)+(-2)(x-2)-4$$
$$= x(x^2+2)-2x$$
$$= x^3$$

Put $\alpha_1=\begin{pmatrix}1\\0\\-1\end{pmatrix}$. Then, $A\alpha_1=0$. Now, put $\beta=\begin{pmatrix}1\\0\\0\end{pmatrix}\notin\langle\alpha_1\rangle$. Then,

$A\beta=\begin{pmatrix}0\\1\\1\end{pmatrix}$, $A^2\beta=\begin{pmatrix}1\\0\\-1\end{pmatrix}\in\langle\alpha_1\rangle$. So put $\alpha_2=\begin{pmatrix}0\\1\\1\end{pmatrix}$.

Let $\alpha_3=\begin{pmatrix}0\\0\\1\end{pmatrix}\notin\langle\alpha_1\rangle\oplus\langle\alpha_2\rangle$. Then, $A\alpha_3=\begin{pmatrix}0\\1\\1\end{pmatrix}\in\langle\alpha_2\rangle$. Now,

for $P=(\alpha_1\ \alpha_2\ \alpha_3)=\begin{pmatrix}1&0&0\\0&1&0\\-1&1&1\end{pmatrix}$, $P^{-1}AP=\begin{pmatrix}0&1&0\\0&0&1\\0&0&0\end{pmatrix}$.

(Actually it is the Jordan-Canonical form)

#6.4.5. If $A^2=A$, then A is diagonizable.

Proof. If $A=0$, then trivial. Assume $A\neq 0$.

Since $A^2-A=0$, the minimal polynomial of A is x^2-x.

So,

$$A\sim\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r=\operatorname{rank}A.$$

#6.4.6. If $T:V\to V$ is diagonalizable,

then for T-invariant subspace W, $T|_W$ is diagonalizable.

Proof Since the minimal polynomial of T splits and separable,

and the minimal polynomial of T_W divides the minimal polynomial of T, so

it is also splits and separable

#6.4.8.

Let $T: V \to V$ be a linear operator.

If every subspace of V is invariant under T, then T is a scalar multiple of Identity.

Proof. Suppose that there is a vector $\alpha \in V$ s.t. $T(\alpha) \notin \langle \alpha \rangle$.

Then, $\langle \alpha \rangle$ is <u>Not</u> T-invariant, so Contradiction.

So, for any $\alpha \in V$, $T(\alpha) = c \cdot \alpha$ for some $c \in F$.

Let $\alpha_1, \alpha_2 \in V$ be independent. Then, $T(\alpha_1) = c_1\alpha_1$ and $T(\alpha_2) = c_2\alpha_2$ for some $c_1, c_2 \in F$.

$$c_1\alpha_1 + c_2\alpha_2 = T(\alpha_1) + T(\alpha_2) = T(\alpha_1 + \alpha_2) = c_3 \cdot (\alpha_1 + \alpha_2)$$

By the independentness, $c_1 = c_3 = c_2$. Now, for fixed $c \in F$, $T(\alpha) = c\alpha$, for all $\alpha \in F$.

#6.4.9.

Let $V = C^0([0,1])$, the space of real-valued continuous map.

Define

$$T: V \to V : f \mapsto \int_0^x f(t)\,dt.$$

1). Determine $\mathbb{R}[x]$ on $[0,1]$ is T-invariant or Not.

2). Determine $D^1[0,1]$ is 〃

3). Determine $H = \{f \in V \mid f(\tfrac{1}{2}) = 0\}$.

Proof. Since the integral of polynomial is polynomial, so T-invariant.

And, 2) is also trivial. But 3) cannot T-invariant,

$$f = (x - \tfrac{1}{2})^2.$$

Comment: 무난 문제2 이게

#6.4.10.

Let $A \in M_{3\times3}(\mathbb{R})$. If A is not similar over $\mathbb{R}$ to a triangular, then A is similar over $\mathbb{C}$ to a diagonal.

Proof. Let $f(t) \in \mathbb{R}[t]$ be the characteristic polynomial of A.

Since $\lim_{t\to\infty} f(t) = \infty$ and $\lim_{t\to-\infty} f(t) = -\infty$, $f(t)$ has at least one zero, by the IVT.

But, since A is not similar to a triangular, or $f(t)$ does not split over $\mathbb{R}$, it cannot have multiple root in $\mathbb{C}$, so separable in $\mathbb{C}$. ⊠

#6.4.11. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$. Then, the char. polynomial is

$f(t) = (t-1)(t-2)$, separable, so it is similar to diagonal.

But A is not already diagonal.

#6.4.12. Let F be an algebraically closed field, and V be a fin. dim. v.s over F. Let $T: V \to V$ be a linear operator on V, and $f(t) \in F[t]$ be a polynomial.

$c \in F$ is an eigenvalue of $f(T)$ if and only if $c = f(\lambda)$ for some eigenvalue λ of T.

Proof. If $c = f(\lambda)$ for some eigenvalue $\lambda \in F$ corresponding a non-zero vector α, then

$f(T)(\alpha) = f(\lambda)\alpha = c\alpha$, so c is eigenvalue of $f(T)$.

To show the converse, denote $f(t) = (t-c_1)\cdots(t-c_n)$, since F is algebraically closed. Now, $c \in F$ is an eigenvalue of $f(T)$, then for some non-zero vector $\alpha \in V$,

$c\alpha = f(T)(\alpha) = (T - c_1 I) \circ \cdots \circ (T - c_n I)(\alpha)$.

Cont'd.

Sec 6.6. Direct-Sum.

#6.6.1. Let V be a fin. dim v.s, and W_1 be any subspace of V.
Prove that there is a subspace W_2 of V such that $V = W_1 \oplus W_2$.
Proof. Let $\{\alpha_1, \cdots, \alpha_k\}$ be the basis for W_1. And, extend this to $\{\alpha_1, \cdots, \alpha_k, \alpha_{k+1}, \cdots, \alpha_n\}$ being the basis for V.
Now, $W_2 = \mathrm{span}_F\{\alpha_{k+1}, \cdots, \alpha_n\}$ is we desired.

#6.6.2. Let V be a fin. dim v.s and $W_1, \cdots, W_k \leq V$ be subspaces s.t

$$V = W_1 + \cdots + W_k \quad \text{and} \quad \dim V = \dim W_1 + \cdots + \dim W_k.$$

Then, $V = W_1 \oplus \cdots \oplus W_k$.
Proof. Let β_i be a basis for W_i, for each $1 \leq i \leq k$. Then,

$\bigcup_{i=1}^{k} \beta_i$ is the spanning set of V,

Moreover, by the assumption, $\bigcup_{i=1}^{k} \beta_i$ is linearly independent.
So, $\{W_1, \cdots, W_k\}$ are independent, obtain the result.

#6.6.3, Find the projection on $\mathbb{R}^2$ onto $\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rangle$ along $\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rangle$.

$$E : \mathbb{R}^2 \to \mathbb{R}^2 ; \alpha \mapsto \frac{1}{3} \cdot \begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} \cdot \alpha.$$

Comment: In General,
let $\{\alpha_1, \cdots, \alpha_k\}$ be a basis for $\mathrm{Im}\, E$ and $\{\alpha_{k+1}, \cdots, \alpha_n\}$ be a basis for $\ker E$.
Then, the invertible Matrix $A = \begin{bmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_n \end{bmatrix}$ satisfies

$$EA = \begin{bmatrix} E\alpha_1 & \cdots & E\alpha_n \end{bmatrix} = \begin{bmatrix} \alpha_1 & \cdots & \alpha_k & 0 & \cdots & 0 \end{bmatrix}$$

Therefore, $E = \begin{bmatrix} \alpha_1 & \cdots & \alpha_k & 0 & \cdots & 0 \end{bmatrix} \cdot A^{-1} = A \cdot \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \cdot A^{-1}$ is we desired.

$$\alpha = a_1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} + a_2 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} \qquad \begin{pmatrix} 1 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a + bc \\ -a + bd \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 6 & -3 \\ -6 & 3 \end{pmatrix}$$

$$A^2 = 3A$$

$$\left(\frac{1}{\sqrt{3}}A\right)^2 = \frac{1}{3} A^2 = \cancel{\frac{4}{3}} A$$

$$\left(\frac{1}{3}A\right)^2 = \frac{1}{9} A^2 = \frac{1}{3} A$$

#6.6.4.
If E_1 and E_2 are Projection such that $\mathrm{Im}\, E_1$ and $\mathrm{Im}\, E_2$ are independent, then E_1+E_2 is Projection. (This statement is false!)

Proof. $(E_1+E_2)^2 = E_1^2+E_2^2+E_1E_2+E_2E_1 = E_1+E_2+E_1E_2+E_2E_1.$

Therefore, E_1+E_2 is projection iff $E_1E_2+E_2E_1=0.$

#6.6.5. If E is projection and f is a polynomial, then $f(E)=aI+bE.$

proof. Let $f(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$. Since E is idempotent,

$$f(E)=(a_n+\cdots+a_1)E+a_0I.$$

6.6.6. False. Counterexample:

$A=\begin{bmatrix}1&1\\0&1\end{bmatrix}$ has the Characteristic polynomial as $f(t)=(t-1)^2$.

But, it is Not diagonalizable, because

$$\ker(A-I)=\mathrm{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\},\ \dim\ 1.$$

#6.6.7. Let E be a Projection. Then, $I-E$ is a projection on $\ker E$ along $\mathrm{Im}\,E$.

Proof. First, $(I-E)^2 = I-E-E+E^2 = I-E$, so $I-E$ is a Projection.

Given $\alpha\in V$, $\alpha\in \mathrm{Im}\,E \Leftrightarrow \alpha=E(\alpha) \Leftrightarrow (I-E)(\alpha)=0 \Leftrightarrow \alpha\in\ker(I-E).$

$\alpha\in\ker E \Leftrightarrow E(\alpha)=0 \Leftrightarrow (I-E)(\alpha)=\alpha \Leftrightarrow \alpha\in \mathrm{Im}(I-E).$

#6.6.8. Let $E_1, \cdots, E_k$ be linear operators on V s.t

$$E_1 + \cdots + E_k = I.$$

a). If $E_i \circ E_j = 0$ for $i \neq j$, then $E_i^2 = E_i$ for each i.

b). In the case $k=2$, prove if $E_1 + E_2 = I$ and $E_1^2 = E_1$, $E_2^2 = E_2$, then $E_1 \circ E_2 = 0$.

Proof.

#6.6.9. Let V be a vector space over $\mathbb{R}$, and E is a projection on V. Then, $I+E$ is invertible.

Proof.
$$\underbrace{(I+E)\circ\left(I - \tfrac{1}{2}\cdot E\right)}_{\substack{\shortparallel \\ (I+E)^{-1}}} = I - \tfrac{1}{2}E + E - \tfrac{1}{2}E^2 = I + \tfrac{1}{2}E - \tfrac{1}{2}E = I.$$

#6.6.10. Let F be a field of characteristic zero, V be a fin. dim space over F. Suppose $E_1, \cdots, E_k$ are projections of V s.t $E_1 + \cdots + E_k = I$. Prove that $E_i \circ E_j = 0$ for $i \neq j$.

Proof. Since every projection on fin. dim space is similar to

$$E_i \sim \begin{bmatrix} I_{r_i} & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r_i = \operatorname{rank} E_i,$$

and trace is invariant under the similar and preserves sum,

$$\dim V = \operatorname{trace} I = \sum \operatorname{trace} E_i = \sum r_i = \sum \dim \operatorname{Im} E_i \quad \cdots (!)$$

So, $V \overset{(*)}{=} \operatorname{Im} E_1 + \cdots + \operatorname{Im} E_k = \operatorname{Im} E_1 \oplus \cdots \oplus \operatorname{Im} E_k$,

$(*)$ is obtained by $\alpha = I(\alpha) = (E_1 + \cdots + E_k)(\alpha) = E_1(\alpha) + \cdots + E_k(\alpha)$ and the direct sum is obtained by $(!)$. Now, given $x \in V$, $x = E_1(x) + \cdots + E_k(x)$. So,

$$i \neq j \Rightarrow E_j(x) \in \operatorname{Ker} E_i \Rightarrow E_i \circ E_j(x) = 0.$$

Sec. 8.1.

#8.1.1. Let V be an Inn. space.

1). For all $\alpha \in V$, $\langle 0, \alpha\rangle = \langle \alpha, 0\rangle = 0$.

2). If $\langle \alpha, \beta\rangle = 0$ for all $\alpha \in V$, then $\beta = 0$.

proof Given $\alpha \in V$, $\langle \alpha, 0\rangle = \langle \alpha, 0\cdot 0\rangle = 0 \cdot \langle \alpha, 0\rangle = 0$.

And, $\langle \beta, \beta\rangle = 0$ implies $\beta = 0$.

#8.1.2 Inner products are closed under the addition and positive scalar multiplication.

#8.1.3. Describe explicitly all inner products on $\mathbb{R}^1$ and $\mathbb{C}^1$.

Proof. Given $\alpha, \beta \in \mathbb{R}$, $\langle \alpha, \beta\rangle = \alpha\beta \cdot \langle 1, 1\rangle$. So, it is determined by the value of $\langle 1, 1\rangle$.

So, the inner product is determined by the value of $\langle 1, 1\rangle > 0$.

Given $a+ib$, $c+id \in \mathbb{C}$,

$$\langle a+ib, c+id\rangle = \langle a, c+id\rangle + \langle ib, c+id\rangle$$
$$= \langle a, c\rangle + \langle a, id\rangle + \langle ib, c\rangle + \langle ib, id\rangle$$

#8.1.5. Let $\langle , \rangle$ be the standard inner product on $\mathbb{R}^2$.

a). Let $\alpha = (1,2)$ and $\beta = (-1,1)$. Find $\gamma \in \mathbb{R}^2$ s.t

$$\langle \alpha, \gamma\rangle = -1 \text{ and } \langle \beta, \gamma\rangle = 3.$$

Let $\gamma = a_1 e_1 + a_2 e_2$. Then, $\langle \alpha, \gamma\rangle = a_1 \cdot \langle \alpha, e_1\rangle + a_2 \cdot \langle \alpha, e_2\rangle = a_1 + 2a_2 = -1$

$\langle \beta, \gamma\rangle = a_1 \cdot \langle \beta, e_1\rangle + a_2 \cdot \langle \beta, e_2\rangle = -a_1 + a_2 = 3$

$a_2 = \frac{2}{3}$, $a_1 = -\frac{17}{3}$.

b). $\alpha = \langle \alpha, e_1\rangle e_1 + \langle \alpha, e_2\rangle e_2$

proof. $\alpha = c_1 e_1 + c_2 e_2$ for some $c_1, c_2 \in \mathbb{R}$. Then,

$$\langle \alpha, e_1\rangle = c_1 \langle e_1, e_1\rangle + 0 = c_1.$$

#8.2.1.

Let $W = \{(1,0,-1,1),(2,3,-1,2)\}^{\perp}$. Find a basis for W.

Proof. Let $W_1 = \mathrm{span}_{\mathbb{R}}\{(1,0,-1,1),(2,3,-1,2)\}$.

Then, $(a,b,c,d) \in W_1^{\perp} \Leftrightarrow \begin{cases} a+0-c+d=0 \\ 2a+3b-c+2d=0 \end{cases} \Rightarrow 3b+c=0$

So, $W^{\perp} = \mathrm{span}\left\{\begin{pmatrix}-1\\0\\0\\1\end{pmatrix}, \begin{pmatrix}-3\\1\\-3\\0\end{pmatrix}\right\}$.

$$\begin{pmatrix}a\\b\\c\\d\end{pmatrix} = \begin{pmatrix}a\\b\\-3b\\d\end{pmatrix} = \begin{pmatrix}c-d\\b\\-3b\\d\end{pmatrix} = \begin{pmatrix}-3b-d\\b\\-3b\\d\end{pmatrix} = b\begin{pmatrix}-3\\1\\-3\\0\end{pmatrix} + d\begin{pmatrix}-1\\0\\0\\1\end{pmatrix}$$

Recall: Row-Reduced Echelon Form.

$$\begin{pmatrix}1&0&-1&1\\2&3&-1&2\end{pmatrix} \to \begin{pmatrix}1&0&-1&1\\0&3&1&0\end{pmatrix} \to \begin{pmatrix}1&0&-1&1\\0&1&\frac{1}{3}&0\end{pmatrix}$$

#8.2.2. Aplly Gram-Schmidt to $\beta_1=(1,0,1)$, $\beta_2=(1,0,-1)$, $\beta_3=(0,3,4)$.

Take $\alpha_1 = \beta_1$,

$$\alpha_2 = \beta_2 - \frac{\langle \beta_2, \alpha_1\rangle}{\|\alpha_1\|^2}\cdot\alpha_1 = (1,0,-1) - 0,$$

$$\alpha_3 = \beta_3 - \frac{\langle \beta_3,\alpha_1\rangle}{\|\alpha_1\|^2}\cdot\alpha_1 - \frac{\langle\beta_3,\alpha_2\rangle}{\|\alpha_2\|^2}\cdot\alpha_2 = (0,3,4) - \frac{4}{2}\cdot(1,0,1) - \frac{-4}{2}\cdot(1,0,-1)$$

$$= (0,3,4) - (2,0,2) + (2,0,-2) = (0,3,0)$$

Now, $\left\{\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\-1\end{pmatrix}, \frac{1}{3}\begin{pmatrix}0\\3\\0\end{pmatrix}\right\}$ is we desired.

#8.2.3. In $\mathbb{C}^3$, fine orthonormal basis for $\mathrm{span}_{\mathbb{C}}\left\{\begin{pmatrix}1\\0\\i\end{pmatrix}, \begin{pmatrix}2\\1\\1+i\end{pmatrix}\right\}$.

Proof. $\begin{pmatrix}2\\1\\1+i\end{pmatrix} - \dfrac{\left\langle \begin{pmatrix}2\\1\\1+i\end{pmatrix}, \begin{pmatrix}1\\0\\i\end{pmatrix}\right\rangle}{2} \cdot \begin{pmatrix}1\\0\\i\end{pmatrix}$ $\quad \frac{1}{2}\cdot\begin{pmatrix}1+i\\2\\1-i\end{pmatrix}$ $\quad 2 \quad 4 \quad 2$

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$= \begin{pmatrix}2\\1\\1+i\end{pmatrix} - \dfrac{2-i+1}{2}\cdot\begin{pmatrix}1\\0\\i\end{pmatrix}$

$= \begin{pmatrix}2\\1\\1+i\end{pmatrix} - \begin{pmatrix}\frac{3-i}{2}\\0\\\frac{1+3i}{2}\end{pmatrix} = \begin{pmatrix}\frac{1+i}{2}\\1\\\frac{1-i}{2}\end{pmatrix}$. So, $\left\{\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\i\end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix}\frac{1+i}{2}\\1\\\frac{1-i}{2}\end{pmatrix}\right\}$.

#8.2.5. Let V be an inn. prod. space, and let $\alpha, \beta \in V$. Then,

$$\alpha = \beta \iff \langle \alpha, \gamma\rangle = \langle \beta, \gamma\rangle \text{ for all } \gamma \in V.$$

proof. $\Rightarrow$ is clear, and $\langle \alpha-\beta, \gamma\rangle = 0$ for all $\gamma \in V$, then $\langle \alpha-\beta, \alpha-\beta\rangle = 0$, so $\alpha = \beta$.

#8.2.6. $E(x_1, x_2) = \dfrac{3x_1+4x_2}{25}\cdot\begin{pmatrix}3\\4\end{pmatrix}$ or $x_1\cdot\begin{pmatrix}\frac{9}{25}\\\frac{12}{25}\end{pmatrix} + x_2\cdot\begin{pmatrix}\frac{12}{25}\\\frac{16}{25}\end{pmatrix}$

$E = \begin{pmatrix}\frac{9}{25} & \frac{12}{25}\\ \frac{12}{25} & \frac{16}{25}\end{pmatrix} \sim \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$

And, $W^\perp = (\mathrm{Im}\,E)^\perp = \ker E$. Clearly, $\ker E = \mathrm{span}\left\{\begin{pmatrix}4\\-3\end{pmatrix}\right\}$ Now, for $A = \begin{pmatrix}3 & 4\\4 & -3\end{pmatrix}$,

$$E = A\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}A^{-1}. \quad \left(A^{-1} = -\frac{1}{25}\cdot\begin{pmatrix}-3 & -4\\-4 & 3\end{pmatrix}\right).$$

#8.2.7. ?

#8.2.8. Find an inner product on $\mathbb{R}^2$ such that $\langle e_1, e_2 \rangle = 2$.

Proof In General, given $x, y \in F^n$ where $F = \mathbb{R}$ or $\mathbb{C}$, $\langle x, y \rangle$ is described by ;
let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an ordered basis for F^n. Construct $A \in M_{n\times n}(F)$ as:

$$A_{i,j} = \langle \alpha_j, \alpha_i \rangle. \quad (\text{so, } A \text{ is Hermitian, } \langle \alpha_j, \alpha_i \rangle = \overline{\langle \alpha_i, \alpha_j \rangle})$$

Then, $x = \sum x_i \alpha_i$ and $y = \sum y_i \alpha_i$,

$$\langle x, y \rangle = \langle \sum x_i \alpha_i, y \rangle = \sum x_i \langle \alpha_i, y \rangle = \sum x_i \langle \alpha_i, \sum y_j \alpha_j \rangle = \sum x_i \cdot \sum \overline{y_j} \cdot \langle \alpha_i, \alpha_j \rangle$$

$$= \sum_{i,j} \overline{y_j} \cdot A_{ji} \cdot x_i = y^* A x.$$

Reversibly, if A is an invertible Hermitian s.t. $x^* A x > 0$ for all $x \neq 0$, then $\langle x, y \rangle = y^* A x$ is an inner product.

Let $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$. $e_2 A e_1 = \left(e_2 \cdot \begin{pmatrix} a \\ b \end{pmatrix}\right) = b$, so, take $b = 2$.

And, $(x_1 \ x_2) A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1 \ x_2) \begin{pmatrix} a x_1 + b x_2 \\ b x_1 + d x_2 \end{pmatrix} = a x_1^2 + 2b x_1 x_2 + d x_2^2 > 0$,

So take $a = 5$, $d = 1$.

#8.2.9. Let $V = P_3(\mathbb{R})$ with equipped an inner product

$$\langle f, g \rangle = \int_0^1 f(t) \cdot g(t) \, dt$$

a). Let $W = \mathbb{R} < V$. Find $W^\perp$.

b). Apply the Gram-Schmidt process to $\{1, x, x^2, x^3\}$.

Proof. Define a map $f \mapsto \int_0^1 f(t) \, dt \left(= \frac{\langle f, 1 \rangle}{\|1\|} \cdot 1 \right)$

Then, it is an orthogonal projection. So, the kernel of this map is $W^\perp$,

$$\{ f \in P_3(\mathbb{R}) \mid \int_0^1 f \, dt = 0 \}.$$

Let $f(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0$. Then,

$$\int_0^1 f \, dt = \frac{1}{4} a_3 t^4 + \frac{1}{3} a_2 t^3 + \frac{1}{2} a_1 t^2 + a_0 t \Big|_0^1 = \frac{1}{4} a_3 + \frac{1}{3} a_2 + \frac{1}{2} a_1 + a_0$$

Therefore, $W^\perp = \{ f \in P_3(\mathbb{R}) \mid \int_0^1 f \, dt = 0 \} = \mathrm{Span}_{\mathbb{R}} \{ x^3 - \frac{1}{4}, x^2 - \frac{1}{3}, x - \frac{1}{2} \}$.

$$x - \frac{\langle x, 1 \rangle}{\|1\|^2} \cdot 1 = x - \int_0^1 t \, dt = x - \frac{1}{2}$$

$$x^2 - \frac{\langle x^2, 1 \rangle}{\|1\|^2} \cdot 1 - \frac{\langle x^2, x - \frac{1}{2} \rangle}{\|x - \frac{1}{2}\|^2} \cdot \left(x - \frac{1}{2} \right) =$$

#8.2.11. Let V be a fin. dim inn. prod space and let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis.
For any $\alpha, \beta \in V$, $\langle \alpha, \beta \rangle = \sum_{k=1}^{n} \langle \alpha, \alpha_k \rangle \cdot \overline{\langle \beta, \alpha_k \rangle}$

Proof. Ask..

$\alpha = \langle \alpha, \alpha_1 \rangle \cdot \alpha_1 + \cdots + \langle \alpha, \alpha_k \rangle \cdot \alpha_k$

$\beta = \langle \beta, \alpha_1 \rangle \cdot \alpha_1 + \cdots + \langle \beta, \alpha_k \rangle \cdot \alpha_k$

So, $\langle \alpha, \beta \rangle = \sum_{i=1}^{n} \langle \alpha, \alpha_i \rangle \cdot \langle \alpha_i, \beta \rangle = \sum_{i=1}^{n} \langle \alpha, \alpha_i \rangle \cdot \left(0 + \cdots + \overline{\langle \beta, \alpha_i \rangle} + \cdots + 0\right)$

$= \sum_{i=1}^{n} \langle \alpha, \alpha_i \rangle \cdot \overline{\langle \beta, \alpha_i \rangle}$

#8.2.12. Let W be a finite-dimensional subspace of V,
and let E be the orthogonal projection of V on W.
Prove that for all $\alpha, \beta \in V$, $\langle E(\alpha), \beta \rangle = \langle \alpha, E(\beta) \rangle$.

Proof. Since E is orthogonal projection, $\operatorname{Im} E = W$, and $\ker E = W^\perp$ with $\operatorname{Im} E \oplus \ker E$.
Given $\alpha, \beta \in V$, there are $\alpha_1, \beta_1 \in \operatorname{Im} E$ and $\alpha_2, \beta_2 \in \ker E$ such that

$\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$

So, $\langle E(\alpha), \beta \rangle = \langle \alpha_1, \beta_1 + \beta_2 \rangle = \langle \alpha_1, \beta_1 \rangle$

and $\langle \alpha, E(\beta) \rangle = \langle \alpha_1 + \alpha_2, \beta_1 \rangle = \langle \alpha_1, \beta_1 \rangle$

Thus, E is a Hermitian.

#8.2.13.

Let S be a subset of an inn. prod space V. Then,

1). $S^{\perp} = \langle S \rangle^{\perp}$ and $S^{\perp}$ is subspace

2). $W \subseteq (W^{\perp})^{\perp}$.

3). If W is fin. dim., then $W = (W^{\perp})^{\perp}$

Proof. Given $\alpha \in V$,

$$\alpha \in S^{\perp} \iff \text{for any } \mu \in S,\ \langle \alpha, \mu \rangle = 0 \underset{\text{linearity of inn. prod.}}{\overset{S \subseteq \langle S \rangle}{\iff}} \text{for any } \mu \in \langle S \rangle,\ \langle \alpha, \mu \rangle = 0 \iff \alpha \in \langle S \rangle^{\perp}.$$

⊣ 1)

Given $\alpha \in W$, $\langle \alpha, \mu \rangle = 0$ for all $\mu \in W^{\perp}$, so $\alpha \in (W^{\perp})^{\perp}$.

But the converse is not true: Let $V = \mathbb{R}[x]$, and define an inn. prod. on V as

$$\langle \cdot , \cdot \rangle : V \times V \to \mathbb{R} ; \left(\sum_{i=1}^{n} a_i x^i , \sum_{i=1}^{m} b_i x^i \right) \mapsto \sum_{i=1}^{n+m} a_i \bar{b}_i .$$

And $W = \langle \{ 1 + x^n \mid n \in \mathbb{N} \} \rangle$. Then, $W \neq V$ and $W^{\perp} = \{0\}$, so $(W^{\perp})^{\perp} = V$. ⊣ 2)

Let W be a fin. dim. Then, there exists an orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$ for W.

Given $\alpha \in (W^{\perp})^{\perp}$, $\langle \alpha, \mu \rangle = 0$ for all $\mu \in W^{\perp}$.

Suppose that $\alpha \notin W$. Then, there is an orthogonal projection $P(\alpha)$ on W.

Namely, $\alpha = P(\alpha) + (\alpha - P(\alpha))$, $\alpha - P(\alpha) \in W^{\perp}$, non-zero by $\alpha \in W$.

But, $\langle \alpha, \alpha - P(\alpha) \rangle = \langle P(\alpha) + (\alpha - P(\alpha)), \alpha - P(\alpha) \rangle = 0 + \| \alpha - P(\alpha) \|^2 > 0$, so

for some $\mu \in W^{\perp}$ ($\mu = \alpha - P(\alpha)$), $\langle \alpha, \mu \rangle \neq 0$. It is contradiction.

#8.2.14. Let V be a fin. dim inn. prod space, and let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis

Let $A = [T]_{\mathcal{B}}$. Then, $A_{i,j} = \langle T(\alpha_j), \alpha_i \rangle$

Proof. Given $1 \le j \le n$, $T(\alpha_j) = \langle T(\alpha_j), \alpha_1 \rangle \cdot \alpha_1 + \cdots + \langle T(\alpha_j), \alpha_n \rangle \cdot \alpha_n$

Therefore,

$$A = \begin{pmatrix} [T(\alpha_1)]_{\mathcal{B}} & \cdots & [T(\alpha_n)]_{\mathcal{B}} \end{pmatrix} = \begin{pmatrix} \langle T(\alpha_1), \alpha_1 \rangle & & \\ \langle T(\alpha_1), \alpha_2 \rangle & \ddots & \\ \vdots & & \ddots \\ \langle T(\alpha_1), \alpha_n \rangle & & \end{pmatrix}.$$

#8.2.15.

Suppose $V = W_1 \oplus W_2$ and $f_i : W_i \times W_i \to F$ $(i=1,2)$ are inner product on W_1 and W_2, respectively, where $F = \mathbb{R}$ or $\mathbb{C}$. Show that there exists a unique inn. Product f on V s.t

a) $W_2 = W_1^{\perp}$

b) $f(\alpha, \beta) = f_k(\alpha, \beta)$ when $\alpha, \beta \in W_k$, $k = 1, 2$

#8.2.16.

Let V be an inner Product space and W is fin.dim Subspace

If a Projection $E: V \to V$ with $\operatorname{Im} E = W$ satisfies

$$\|E(\alpha)\| \leq \|\alpha\| \quad \text{for all } \alpha \in V,$$

then E is orthogonal Projection on W.

#8.2.17. Let $V = \{f : [-1,1] \to \mathbb{R} \mid f \text{ is Continuous}\}$ with equipped an inner Product

$$\langle f, g \rangle = \int_{-1}^{1} f(t) \cdot g(t)\, dt$$

Let W be the subspace of odd functions. Find $W^\perp$.

Proof. $f \in W^\perp \iff$ for any odd function g, $\int_{-1}^{1} f(t)\, g(t)\, dt = 0$

Since for any even function g and any odd f, fg is odd, so $\int_{-1}^{1} fg = 0$.

That is, $W^\perp$ Contains all even function. Moreover, for any $f \in V$,

$$f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd}}$$

So, for subset W' of V consisting by all even functions,

$$V = W + W', \quad \text{and} \quad W' \subseteq W^\perp.$$

Now, clearly $W \cap W' = \{0\}$, $V = W \oplus W'$.

If $W' \neq W^\perp$, then, for some $h \in W^\perp$, $h \notin W'$. Again, h can be decomposed as

$$h(x) = \underbrace{\frac{h(x) + h(-x)}{2}}_{\in W'} + \underbrace{\frac{h(x) - h(-x)}{2}}_{\in W}$$

Since $W' \subset W^\perp$, $h(x) - \frac{h(x)+h(-x)}{2} \in W^\perp$, and $\in W$.

But, this implies that it is contained in $W \cap W^\perp$, so it is zero:

If $x \in W \cap W^\perp$, then $\langle x, x \rangle = 0$, or $x = 0$.

Thus, $h(x) - \frac{h(x)+h(-x)}{2} = \frac{h(x)-h(-x)}{2} = 0$, So $h(x) = h(-x)$, even. Contradicts to $h \notin W'$.

Now, we can construct a projection of V onto W as:

$$P : V \to V : f(x) \mapsto \frac{f(x) - f(-x)}{2}$$

8.3 Adjoint.

#8.3.1. In $\mathbb{C}^2$, equipped the standard inn. prod, let T be the linear operator defined by $Te_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$, $Te_2 = \begin{pmatrix} i \\ -1 \end{pmatrix}$.

Find a formula $T^*(x_1, x_2)$.

Proof Since $\{e_1, e_2\}$ is orthonormal basis for $\mathbb{C}^2$,

$$[T^*]_\beta = [T]_\beta^* = \begin{pmatrix} 1 & i \\ -2 & -1 \end{pmatrix}^* = \begin{pmatrix} 1 & -2 \\ -i & -1 \end{pmatrix}.$$

So, $$T^*(x_1, x_2) = \begin{pmatrix} 1 & -2 \\ -i & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 - 2x_2 \\ -ix_1 - x_2 \end{pmatrix}.$$

#8.3.2. $[T]_\beta = \begin{pmatrix} 1+i & i \\ 2 & i \end{pmatrix}$, $[T^*]_\beta = \begin{pmatrix} 1-i & 2 \\ -i & -i \end{pmatrix}$

Commute?

$$T \cdot T^* = \begin{pmatrix} 3 & \\ & \end{pmatrix} \neq T^* T = \begin{pmatrix} 6 & \\ & \end{pmatrix}$$

No!

#8.3.3. $T = \begin{bmatrix} -1 & -i & 1 \\ -i & 1 & i \\ 1 & i & -1 \end{bmatrix}$, $T^* = \begin{bmatrix} -1 & i & 1 \\ i & 1 & -i \\ 1 & -i & -1 \end{bmatrix}$

#8.3.4

Let V be a fin. dim inn. prod space, and $T: V \to V$ be a linear operator. Then,

$$\operatorname{Im} T^* = (\ker T)^\perp.$$

Proof. Let $T^*(\alpha) \in \operatorname{Im} T^*$ be given. Then, for any $\beta \in \ker T$,

$$\langle T^*(\alpha), \beta \rangle = \langle \alpha, T(\beta) \rangle = \langle \alpha, 0 \rangle = 0, \quad \text{so } \operatorname{Im} T^* \subseteq (\ker T)^\perp.$$

Conversely, since V is finite dimensional,

$$((\ker T)^\perp)^\perp = \ker T$$

and $\alpha \in \ker T \iff$ for all $\beta \in V$, $0 = \langle 0, \beta \rangle = \langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle$

$\iff \alpha \in (\operatorname{Im} T^*)^\perp$.

So, $\ker T = (\operatorname{Im} T^*)^\perp$, so $(\ker T)^\perp = ((\operatorname{Im} T^*)^\perp)^\perp = \operatorname{Im} T^*$.

#8.3.5.

#8.3.6. Let V be an inner product space, and β, γ be fixed vectors in V.
Show that
$$T : V \to V : \alpha \mapsto \langle \alpha, \beta \rangle \cdot \gamma$$
is a linear operator, and it has an adjoint. Describe explicitly the adjoint.

Proof. 1). T is linear. Given $\alpha_1, \alpha_2 \in V$ and $c \in F$,
$$T(c\alpha_1 + \alpha_2) = \langle c\alpha_1 + \alpha_2, \beta \rangle \cdot \gamma = (c \cdot \langle \alpha_1, \beta \rangle + \langle \alpha_2, \beta \rangle) \cdot \gamma = c \cdot T(\alpha_1) + T(\alpha_2).$$

2). T has an adjoint.
Given $\alpha_1, \alpha_2 \in V$,
$$\langle T(\alpha_1), \alpha_2 \rangle = \langle \langle \alpha_1, \beta \rangle \cdot \gamma, \alpha_2 \rangle = \langle \alpha_1, \beta \rangle \cdot \langle \gamma, \alpha_2 \rangle = \langle \gamma, \alpha_2 \rangle \cdot \langle \alpha_1, \beta \rangle = \langle \alpha_1, \langle \alpha_2, \gamma \rangle \cdot \beta \rangle$$
So, $T^*(\alpha) = \langle \alpha, \gamma \rangle \cdot \beta$ is the adjoint of T.

Particularly, let $\mathbb{C}^n$ with standard inn. prod. Put
$$\beta = (y_1, \cdots, y_n) \quad \text{and} \quad \gamma = (x_1, \cdots, x_n)$$
$$T(e_i) = \langle e_i, \beta \rangle \cdot \gamma = \bar{y}_i \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$
Then, for standard basis $\mathcal{B}$ for $\mathbb{C}^n$,
$$[T]_{\mathcal{B}} = \begin{pmatrix} [T(e_1)]_{\mathcal{B}} & \cdots & [T(e_n)]_{\mathcal{B}} \end{pmatrix} = \begin{pmatrix} x_1\bar{y}_1 & \cdots & \cdots \\ x_2\bar{y}_1 & \ddots & \\ \vdots & & \ddots \\ x_n\bar{y}_1 & \cdots & x_n\bar{y}_n \end{pmatrix} = (x_i\bar{y}_j)_{i,j}$$
The rank of $[T]_{\mathcal{B}}$ is exactly 1, if neither β nor γ are non-zero.

Note: It is element of the singular value decomposition.

#8.3.8.

3, 4 Let $V = P_3(\mathbb{R})$ with inn. prod $\langle f, g \rangle = \int_0^1 f(x) g(x)\, dx$

For a fixed $t \in \mathbb{R}$, find $g_t \in V$ s.t.

$$\langle f, g_t \rangle = f(t) \text{ for all } f \in V.$$

And, for differentiation D, find D^*.

Proof. Since $V \to \mathbb{R} : f \mapsto f(t)$ is a linear functional,

the Riesz-representation Thm gives that for some $g_t \in V$, $\langle \cdot, g_t \rangle = f \mapsto f(t)$.

More specifically, take $\beta = \{\alpha_1, \cdots, \alpha_4\}$ as orthonormal basis for V.

Then, for $g_t = \overline{\alpha_1(t)} \cdot \alpha_1 + \overline{\alpha_2(t)} \cdot \alpha_2 + \overline{\alpha_3(t)} \cdot \alpha_3 + \overline{\alpha_4(t)} \cdot \alpha_4$,

$\langle f, g_t \rangle = \sum \alpha_i(t) \cdot \langle f, \alpha_i \rangle = f(t)$.

Construct an orthonormal basis as; for $\beta = \{1, x, x^2, x^3\}$

$\alpha_1 = 1$.

$$x^4 - x^3 + \frac{1}{6}x^2$$
$$-x^3 + x^2 - \frac{1}{6}x$$
$$+\frac{1}{6}x^2 - \frac{1}{6}x + \frac{1}{36}$$

$\alpha_2 = x - \dfrac{\int_0^1 x \cdot 1\, dx}{\|1\|^2} \cdot 1 = x - \frac{1}{2}$.

$\alpha_3 = x^2 - \dfrac{\int_0^1 x^2 \cdot 1\, dx}{\|1\|^2} \cdot 1 - \dfrac{\int_0^1 x^2 \cdot (x - \frac{1}{2})\, dx}{\|x - \frac{1}{2}\|^2} \cdot (x - \frac{1}{2})$ $= x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36}$

$= x^2 - \frac{1}{3} - (x - \frac{1}{2}) = x^2 - x + \frac{1}{6}$. $\frac{3}{40}$ $\frac{1}{120}$.

$\alpha_4 = x^3 - \dfrac{\int_0^1 x^3 \cdot 1\, dx}{\|1\|^2} \cdot 1 - \dfrac{\int_0^1 x^3 \cdot (x - \frac{1}{2})\, dx}{\|x - \frac{1}{2}\|^2 = \frac{1}{12}} (x - \frac{1}{2}) - \dfrac{\int_0^1 x^3 (x^2 - x + \frac{1}{6})}{\|(x^2 - x + \frac{1}{6})\|^2 = \frac{1}{180}} \cdot (x^2 - x + \frac{1}{6})$

$= x^3 - \frac{3}{2}x^2 - \frac{12}{5}x - \frac{1}{20}$. And, processing Gram-Schmidt process we can obtain that

And, find the adjoint of differentiation: $\langle Df, g \rangle = \langle f, D^* g \rangle$, so

$\langle Df, g \rangle = \int_0^1 Df(x)\, g(x)\, dx = \int_0^1 D(f(x) g(x))\, dx - \int_0^1 f(x)\, Dg(x)\, dx = \langle D(fg), 1 \rangle - \langle f, Dg \rangle$

It equals to $\langle f, D^* g \rangle$, so

$$\langle f, D^* g + Dg \rangle = \langle f, (D^* + D)(g) \rangle = \langle D(fg), 1 \rangle = f(1)g(1) - f(0)g(0)$$

Let $g = 1$.

#8.3.10. Let $V = M_{n\times n}(\mathbb{C})$, with an inner product

$$\langle A, B\rangle = \mathrm{trace}(AB^*).$$

Let P be a fixed invertible Matrix in V, and $T_P(A) = P^{-1}AP$. Find T_P^*.

Proof. Given $A, B \in V$,

$$\langle T_P(A), B\rangle = \langle P^{-1}AP, B\rangle = tr(P^{-1}APB^*) = tr(APB^*P^{-1}) = \langle A, (PB^*P^{-1})^*\rangle$$
$$= \langle A, (P^*)^{-1}BP^*\rangle = \langle A, T_{P^*}(B)\rangle$$

So, $(T_P)^* = T_{P^*}$.

#8.3.11 Let V be a fin.dim inn. Prod space, and E be an idempotent linear map on V. Then, E is Hermitian iff E is Normal.

Proof. If Hermitian, then clearly normal. (Not necessary idempotent).
If E is Normal, that is, $EE^* = E^*E$, then given $\alpha, \beta \in V$,
$\langle EE^*(\alpha), \beta\rangle = \langle E^*(\alpha), E^*(\beta)\rangle \underset{\text{normal}}{=} \langle E^*E(\alpha), \beta\rangle = \langle E(\alpha), E(\beta)\rangle$
So, $\|E^*(x)\| = 0 \Leftrightarrow \|E(x)\| = 0$, or $\ker E^* = \ker E$.
Since E is an idempotent, for any $x \in V$, $x - E(x) \in \ker E = \ker E^*$,
So $E^*(x - E(x)) = E^*(x) - (E^*\circ E)(x) = 0$, thus $E^*(x) = (E^*\circ E)(x) = (E\circ E^*)(x)$.
That is, $E^* = E^*\circ E$. Since $(E^*\circ E)^* = E^*\circ E^{**} = E^*\circ E$, $E^* = E$. ∎

#8.3.12. Let V be a fin.dim Complex inn. Prod space, and let T be a linear operator on V. Then, T is Hermitian if and only if for all $\alpha \in V$, $\langle T(\alpha), \alpha\rangle \in \mathbb{R}$.

Proof. If T is hermitian, given $\alpha \in V$, $\langle T(\alpha), \alpha\rangle = \langle \alpha, T^*(\alpha)\rangle = \langle \alpha, T(\alpha)\rangle = \overline{\langle T(\alpha), \alpha\rangle}$,
so it is real-valued. To show the Converse, we will show that; $\langle T(\alpha), \alpha\rangle = 0$ for all $\alpha \in V$, then $T = T_0$.

#8.3.12. Let V be a fin. dim Complex inn. Prod space, and let T be a linear operator on V. Then, T is Hermition if and only if for all $\alpha \in V$, $\langle T(\alpha), \alpha \rangle \in \mathbb{R}$.

Proof. If T is hermitian, given $\alpha \in V$, $\langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \overline{\langle T(\alpha), \alpha \rangle}$, so it is real-valued. To show the Converse, we will show that; $\langle T(\alpha), \alpha \rangle = 0$ for all $\alpha \in V$, then $T = T_0$.

Given $x, y \in V$,

$$0 = \langle T(x+y), x+y \rangle = \langle T(x), x \rangle + \langle T(y), y \rangle + \langle T(x), y \rangle + \langle T(y), x \rangle = \langle T(x), y \rangle + \langle T(y), x \rangle$$

and

$$0 = \langle T(x+iy), x+iy \rangle = \langle T(x), iy \rangle + \langle T(iy), x \rangle = -i \cdot \langle T(x), y \rangle + i \cdot \langle T(y), x \rangle$$

Thus, $0 = -\langle T(x), y \rangle + \langle T(y), x \rangle$

Combining the above the results, then $\langle T(x), y \rangle = 0$, for all $x, y \in V$.

Take $y = T(x)$, then $\|T(x)\|^2 = 0$ for all x, so $T = T_0$.

Now, observe that $T = T^* \iff T - T^* \equiv 0$.

If for all $\alpha \in V$, $\langle T(\alpha), \alpha \rangle \in \mathbb{R}$,

$$\langle T(\alpha), \alpha \rangle = \overline{\langle \alpha, T(\alpha) \rangle} = \langle \alpha, T(\alpha) \rangle = \langle T^*(\alpha), \alpha \rangle$$

So, $\langle (T - T^*)(\alpha), \alpha \rangle = 0$, it implies $T - T^* \equiv 0$, or T is Hermitian.

「Section. 8.4 Unitary operator.

#1. Ask a toddler

#2. Let $V = M_{n\times n}(\mathbb{C})$ with an inner product

$$\langle A, B\rangle = tr(AB^*).$$

For each $M \in V$, define $T_M(A) = MA$. Show that

T_M is unitary if and only if M is unitary.

Proof. T_M is unitary $\Leftrightarrow$ for all A and B in V, $\langle MA, MB\rangle = \langle A, B\rangle$.

$\Leftrightarrow$ ", $tr(MAB^*M^*) = tr(AB^*)$

Since $tr(MAB^*M^*) = tr(M^*MAB^*)$, if M is unitary,

$= tr(AB^*)$, so T_M is unitary.

And,

#3. Let $V=\mathbb{C}$ be the vector space over $\mathbb{R}$.

a). $\langle \alpha, \beta\rangle = \mathrm{Re}(\alpha\bar{\beta})$ is an inner product on V.

Proof. Given $\alpha, \beta, \gamma \in \mathbb{C}$ and $c \in \mathbb{R}$,

$$\langle c\alpha+\beta, \gamma\rangle = \mathrm{Re}([c\alpha+\beta]\cdot\bar{\gamma}) = \mathrm{Re}(c\cdot\alpha\cdot\bar{\gamma}+\beta\bar{\gamma}) = c\cdot\mathrm{Re}(\alpha\cdot\bar{\gamma}) + \mathrm{Re}(\beta\bar{\gamma})$$

$$= c\cdot\langle\alpha,\gamma\rangle + \langle\beta,\gamma\rangle. \text{ linearity}$$

$$\overline{\langle \alpha,\beta\rangle} = \overline{\mathrm{Re}(\alpha\bar{\beta})} = \mathrm{Re}(\alpha\bar{\beta}) = \mathrm{Re}(\bar{\beta}\alpha) = \mathrm{Re}(\overline{\beta\bar{\alpha}}) = \mathrm{Re}(\beta\bar{\alpha}) = \langle\beta,\alpha\rangle \text{ Condi.}$$

$$\langle\alpha,\alpha\rangle = \mathrm{Re}(\alpha\bar{\alpha}) = \mathrm{Re}(|\alpha|^2) > 0 \text{ iff } \alpha\neq 0. \text{ positive}$$

b). Find inner product preserved isomorphism onto $\mathbb{R}^2$ with standard inn. prod.

Proof. Clearly, $\mathcal{B}_2 = \{e_1, e_2\}$ is orthonormal basis for $\mathbb{R}^2$.

And, $\mathcal{B}_1 = \{1, i\}$ is an orthonormal basis for $\mathbb{C}$ over $\mathbb{R}$,

$\langle 1, 1\rangle = 1$, $\langle i, i\rangle = \mathrm{Re}(i\cdot -i) = \mathrm{Re}(1) = 1$, $\langle 1, i\rangle = \mathrm{Re}(-i) = 0$

Now, define $T: \mathbb{C}\to\mathbb{R}^2$ s.t. $T(1) = e_1$, $T(i) = e_2$.

That is, $[T]_{\mathcal{B}_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1\end{bmatrix}$, or $T(a+bi) = \begin{pmatrix} a \\ b\end{pmatrix}$.

c). For each $\gamma\in V$, define $M_\gamma: V\to V: \alpha\mapsto\gamma\alpha$.

Show that $(M_\gamma)^* = M_{\bar{\gamma}}$

Proof. Given $\alpha, \beta\in V$,

$$\langle M_\gamma(\alpha), \beta\rangle = \langle\gamma\alpha, \beta\rangle = \mathrm{Re}(\gamma\alpha\bar{\beta}) = \langle\alpha, \beta\bar{\gamma}\rangle = \langle\alpha, M^*_\gamma(\beta)\rangle$$

Since α and β were arbitrary, $M^*_\gamma(\beta) = \beta\bar{\gamma} = M_{\bar{\gamma}}(\beta)$.

+d),

So, $M_\gamma = M^*_\gamma \iff \gamma\alpha = \bar{\gamma}\alpha$ for all $\alpha\in\mathbb{C}$, so $\gamma=\bar{\gamma}$, or γ be real.

⌈Normal and Hermitian.

⌈Lemma. Let V be an inn. Prod space and T is a linear operator.
(If exists) Every eigenvalue of T is real.
Moreover, eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Given eigenvalue c of T corresponding to α,
$$c\cdot\langle\alpha,\alpha\rangle = \langle T(\alpha),\alpha\rangle = \langle\alpha, T^*(\alpha)\rangle = \langle\alpha, T(\alpha)\rangle = \langle\alpha, c\cdot\alpha\rangle = \bar{c}\cdot\langle\alpha,\alpha\rangle$$
Since α is non-zero, $c=\bar{c}$, or c is a real number.
And, take α_1,α_2 be eigenvectors corresponding to c_1, c_2, with $c_1\neq c_2$.
Then,
$$c_1\cdot\langle\alpha_1,\alpha_2\rangle = \langle c_1\alpha_1,\alpha_2\rangle = \langle T(\alpha_1),\alpha_2\rangle = \langle\alpha_1, T^*(\alpha_2)\rangle = \langle\alpha_1, T(\alpha_2)\rangle = \langle\alpha_1, c_2\alpha_2\rangle = \bar{c}_2\cdot\langle\alpha_1,\alpha_2\rangle$$
Since c_2 must be real, $\bar{c}_2 = c_2$. Now, since $c_1\neq c_2$,
$$(c_1-c_2)\cdot\langle\alpha_1,\alpha_2\rangle = 0 \text{ implies } \langle\alpha_1,\alpha_2\rangle = 0.$$

⌈Thm. In finite-dimensional inn. prod space,
Every Hermitian operator has an eigenvector.

Proof. If the given field is $\mathbb{C}$, then the charateristic polynomial must be a root in $\mathbb{C}$ by the Fundamental Theorem of Algebra. So it has an eigenvector.
Now, assume that the given field is the $\mathbb{R}$.
Let T be a Hermitian on V over $\mathbb{R}$, and β be an orthonormal basis for V.
Let $A=[T]_\beta$. Since $T=T^*=T^t$, $A=A^*=A^t$. Define an inner product on $\mathbb{C}^n$ as:
$$\langle X,Y\rangle \overset{\text{def}}{=} Y^*X$$
Then, a linear operator $L_A:\mathbb{C}^n\to\mathbb{C}^n : X\mapsto AX$ is Hermitian, because
$$\langle L_A(X),Y\rangle = \langle X, L_A^*(Y)\rangle \text{ and } \langle AX,Y\rangle = Y^*AX = \langle X, A^*Y\rangle = \langle X, AY\rangle.$$
By the FTA, L_A has an eigenvalue, and that must be real, by the above Lemma.
Now, $L_A(X) = AX = cX$ for some real number c and a nonzero vector X in $\mathbb{C}^n$.
or $\det(cI-A)=0$, $A-cI$ is not invertible, and $A-cI$ has only real-entries,
so $(A-cI)X=0$ has a non-zero real solution X, so proved.

「Sec. 8.5.

#1. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$\det(A-tI) = (t-1)^2 - 1 = t(t-2)$, so

let $P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, then $P^{-1} = \frac{1}{1} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

$$P^{-1}AP = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$$

$\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$

$$\det(tI - A) = (t^2 - \cos^2\theta) - \sin^2\theta = t^2 - 1 = (t-1)(t+1)$$

$$\begin{bmatrix} \cos\theta + 1 & \sin\theta \\ \sin\theta & -\cos\theta + 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{pmatrix} x\cos\theta + y\sin\theta + x \\ x\sin\theta - y\cos\theta + y \end{pmatrix} = 0.$$

$$\begin{cases} x(\cos\theta + 1) + y\sin\theta = 0 \\ y(1 - \cos\theta) + x\sin\theta = 0 \end{cases}$$

$y = \frac{\sin\theta}{1 - \cos\theta}$, $x \cdot (\cos\theta + 1) = \sin\theta \cdot \frac{\sin\theta}{\cos\theta - 1} = \frac{1 - \cos^2\theta}{\cos\theta - 1} = -(1 + \cos\theta)$

so $x = -1$. $\begin{pmatrix} \cos\theta - 1 \\ \sin\theta \end{pmatrix}$

$P = \begin{pmatrix} \cos\theta - 1 \\ \sin\theta \end{pmatrix}$

$\sin^2\theta + \cos^2\theta - 2\cos\theta + 1$

$= 2 - 2\cos\theta$

#3. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$

Since the charateristic polynomial is invariant by similar,

$$\det(tI - A) = \begin{vmatrix} t-1 & -2 & -3 \\ -2 & t-3 & -4 \\ -3 & -4 & t-5 \end{vmatrix}$$